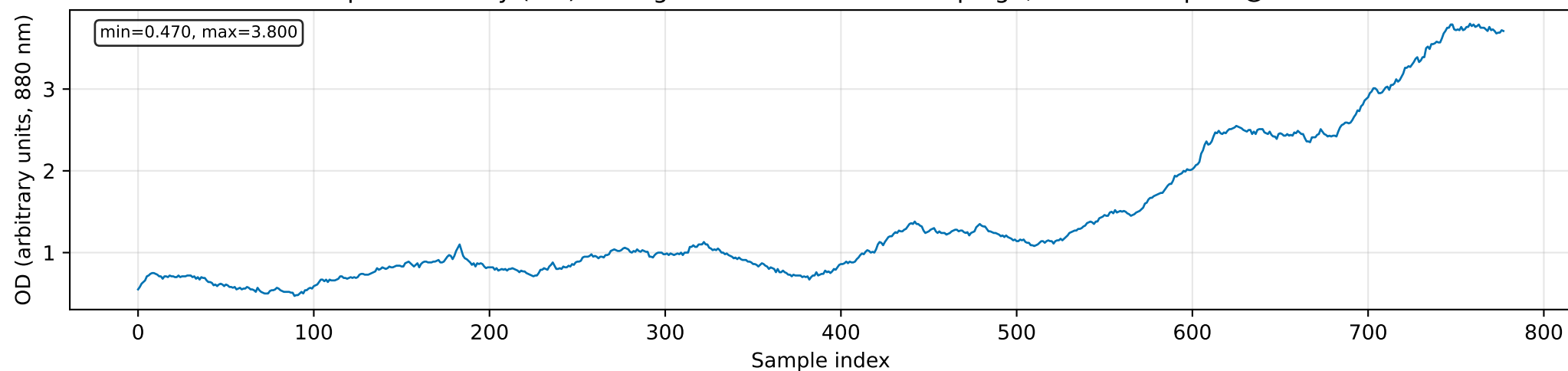
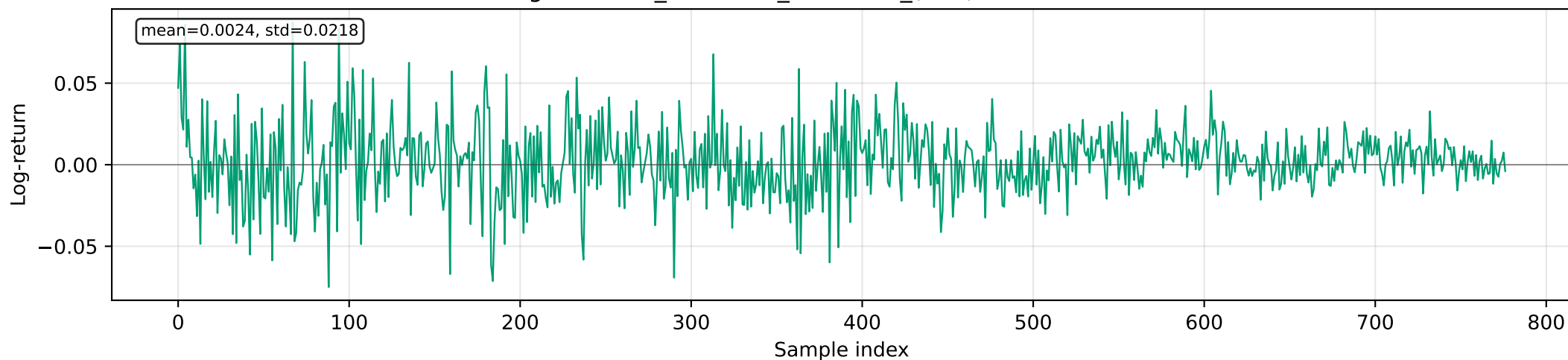


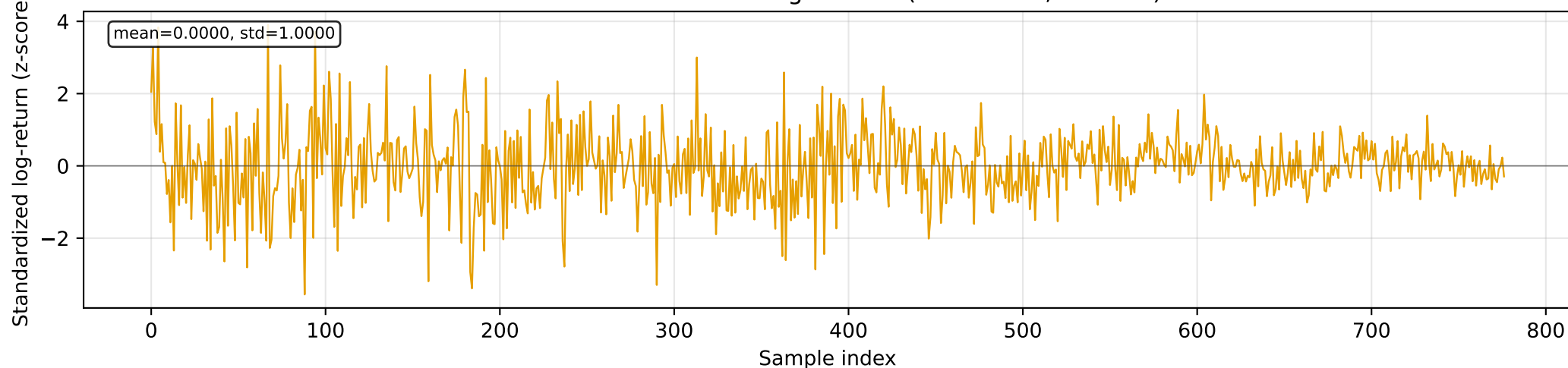
Preprocessing pipeline — four stages from raw OD to model-ready input
Panel 1: Raw optical density (OD) — single LUCY cultivation campaign, n=778 samples @ 10-min intervals



Panel 2: Log-returns $r_t = \ln OD_t - \ln OD_{(t-1)}$ — first-difference, n=777



Panel 3: Standardized log-returns (zero mean, unit std)



Panel 4: Inverse Lambert W heavy-tail correction ($\delta=0.1469$) + rescale to $[-1, 1]$ — model training input

